



SETU - Curiosity to Creativity

(Student's Engagement for Transformative Upskilling)

A project based learning initiative by Zoho Nagpur

SOFTWARE DOMAIN

Query / Doubt Clearance:

- For any query related to any project:
 - Fill out the following form to start communication regarding any query: <https://zfrmz.in/HoTJfl4wnk9JUBgmuXn0>
 - Once, we receive your query, we will get back to you over mail. Further communication will be continued over the same mail chain.
- For any non technical Query:
 - For any non-technical query, you can mail it to setu@dobox.org
 - Mention your name, branch and college properly in the mail.

Project List:

- Following is the project list. You may choose any one from the following list.

Sr No.	Project	Project Type	Description	Reference Links
1	Develop a File Server	File Sharing protocols	Develop a cross-platform, GUI-based file server that runs on Linux and Windows, with optional macOS support. It allows authenticated users to securely upload, download, and manage files over a network. The system focuses on reliable multi-user access, core file operations, and strong system design rather than visual aesthetics.	File Server Explained
				Network-file-system
				Server-Message-Block-Protocol
2	LustreFS on Network Block Storage	Distributed Storage	Design and implement a CLI-based storage management tool on Linux that exports and imports block devices using iSCSI and uses the imported devices to deploy a Lustre filesystem. Students are expected to validate the setup through basic operations, performance benchmarking, and exploration of failure or incorrect configurations.	Lustre , iSCSI
3	Distributed Linux Fleet Management	System Orchestration	Design and implement a Web-based Linux Fleet Management system that can onboard and manage multiple Linux nodes remotely, and collect, store, and visualise system metrics using industry-standard monitoring tools (Prometheus and Grafana). The project prioritises correctness, system design, observability principles, and understanding of Linux internals. UI aesthetics are not a priority.	Prometheus
				Grafana

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4	Secure Shared Storage using LDAP + Kerberos + NFS	Data Security	Design and deploy a secure, centrally managed shared storage system where: User identities are managed centrally Authentication is strong and ticket-based (not password-based file access) Multiple machines access the same filesystem consistently and securely The system behaves correctly during failures	NFS
				LDAP
				Kerberos
5	Usage Metering & Billing Engine for Object Storage	Cloud Infrastructure	Design and build a multi-tenant object storage service that behaves like a cloud storage product, where: Multiple users can upload, access, and manage objects Each user's data is strictly isolated Real usage is measured automatically A usage-based bill is generated per user	IBM Billing
				IBM S3 API
				Storage Types
6	BeeGFS (benchmarking & integration)	Filesystem	Benchmark BeeGFS performance and optionally integrate with tools like Spark or HBase.	https://en.wikipedia.org/wiki/BeeGFS

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7	Containerized HBase with Hadoop	Storage System	Deploy HBase and Hadoop in Docker containers, simulate workloads, and evaluate performance.	https://hbase.apache.org/ , https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-hdfs/HdfsDesign.html
8	NetBoot Deployment System	Network Booting	A network-based boot system that enables client machines to load operating systems without local storage using PXE technology.	https://en.wikipedia.org/wiki/Network_booting
9	Fuse and its protocols	Filesystem	Build a custom FUSE-based filesystem or protocol extension (e.g., encryption, compression, cloud mount).	https://en.wikipedia.org/wiki/Filesystem_in_Userspace
10	Baseline kernel OS	Linux	Create a minimal Linux kernel with core components like drivers, memory management, and basic shell.	https://kernel.org/
11	LLVM Compiler for Cortex-M0	LLVM	Write LLVM front-end code for ARM Cortex-M0	https://llvm.org/
12	Storage Tiering	Storage System	Develop an end-to-end auto tiering solution considering factors like performance, availability and storage costs. An auto tiering solution comprises monitoring and classifying data based on its (access pattern, type, size, etc) and then migrating data to appropriate storage tier.	https://www.techtarget.com/searchstorage/definition/tiered-storage
				https://www.usenix.org/system/files/conference/osdi14/osdi14-paper-muralidhar.pdf
				https://aws.amazon.com/s3/stora

Sr No.	Project	Project Type	Description	Reference Links
				ge-classes/intelligent-tiering/ https://cloud.google.com/storage/docs/storage-classes https://learn.microsoft.com/en-us/azure/storage/blobs/access-tiers-overview
13	Yocto - Raspberrypi, Beaglebone	Firmware	Build a yocto image for Raspberrypi Board/ BeagleBone Black and add an application to control and monitor hardware components (example Sound Sensor or Humidity Sensors) Note: Use any open source build system like meson or cmake.	https://docs.yoctoproject.org/
14	OpenBMC image for Raspberry Pi	Firmware	Build an open-source OpenBMC Baseboard Management Controllers (BMC) Firmware binary for RaspberryPi machine. Also add Temp Sensor(LM75) and display the sensor data under Hardware Status of OpenBMC WebUI. 2. Change the OpenBMC WebUI logo-header to your respective college logo.	https://www.openbmc.org/
15	OpenBMC image for Raspberry Pi	Firmware	Build an open-source OpenBMC Baseboard Management Controllers (BMC) Firmware binary for RaspberryPi machine. Also add an application to control (read and write) value of a GPIO through BMC Console. Note: Use any open source build system like meson or cmake. 2. Change the OpenBMC WebUI logo-header to your respective college logo.	https://www.openbmc.org/
16	OpenBMC image for Raspberry Pi	Firmware	Build an OpenBMC Baseboard Management Controllers (BMC) Firmware binary for RaspberryPi machine. Add an application to monitor noise levels using a sound sensor and visualize the measured data as real-time graphs in a WebUI	https://www.openbmc.org/
17	OpenBMC image for Raspberry Pi	Firmware	Build an open-source OpenBMC Baseboard Management Controllers (BMC) Firmware binary for RaspberryPi machine. Also add a GPIO-based LED control service to dbus-sensors, accessible via Redfish API and make it visible in the OpenBMC WebUI.	https://www.openbmc.org/ https://www.dmtf.org/standards/redfish

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			Note: Use any open source build system like meson or cmake.	
18	RTOS - Zephyr OS	Firmware	Write an application to read value from temperature sensor and indicate it via LEDs using State Machine for Zephyr OS. Write a State machine code for reading temperature value such that, value above than threshold will indicate Red LED blink and value below the threshold will indicate Yellow LED blink.	https://www.zephyrproject.org/
19	Yocto - TI AM62X	Firmware	Build a yocto image for TI AM62X and create an application for PWM-Based Fan Control Mechanism based on tempearture sensor value .	https://docs.yoctoproject.org/ https://www.ti.com/tool/SK-AM62
20	Minimal Bootloader for ARM Core	Firmware	Write a minimal bootloader (in C or Rust) that directly runs on the ARM core, loads a kernel image from SD, and jumps to it.	
21	Coreboot and Slimbootloader for QEMU	Firmware	Build and test Coreboot and SeaBIOS on Quick Emulator (QEMU). Along with it, build and test Slimbootloader with OsLoader for QEMU.	https://slimbootloader.github.io/ https://www.coreboot.org/
22	UEFI Boot Menu on Raspberry Pi using EDK-II	Firmware	Building a Custom UEFI Boot Menu on a Raspberry Pi (Pi 3 or 4) using EDK II (Tianocore). This boot menu will let users select from multiple operating systems (Linux, RTOS, hobby OS) on boot.	https://slimbootloader.github.io/
23	Network-based Multi-boot System using SBL	Firmware	Design and implement a network-based multi-boot system using Slim Bootloader (SBL) that enables users to choose between multiple operating system images (e.g., Ubuntu, Alpine, custom Linux) at boot time via a menu-driven interface. The system should leverage PXE-like netbooting by retrieving boot images (kernel and initrd) from a central TFTP server, enabling diskless booting, rapid OS switching, and simplified OS image management.	https://slimbootloader.github.io/